12. Develop an Android application to fetch and parse JSON data from a web

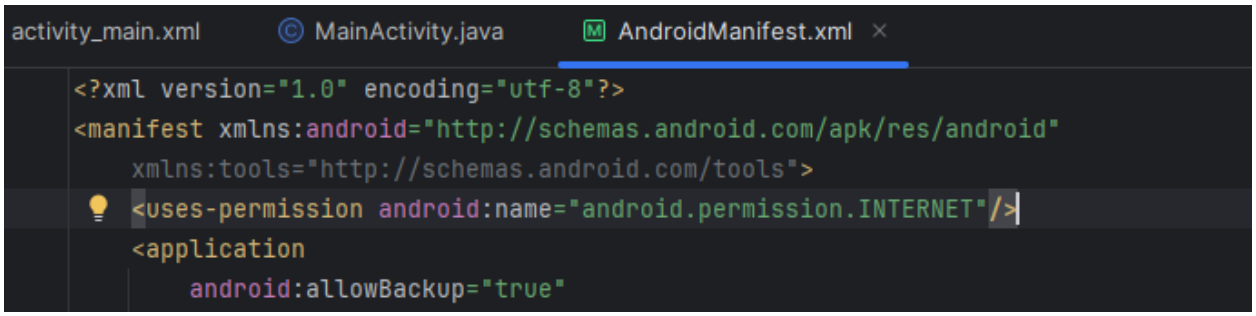
PROJECTNAME: JSONDEMO

Add Internet Permission.

AndroidManifest.xml

Add following above: <application> tag

<uses-permission android:name="android.permission.INTERNET"/>



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">



   <Button

       android:id="@+id/btnFetch"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"
```

```
        android:text="Fetch JSON Data" />


    <TextView
        android:id="@+id/txtResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:textSize="18sp"
        android:textColor="#000000" />


</LinearLayout>
```

MainActivity.java

package com.example.jsondemo;

```
package com.example.jsondemo;


import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.net.HttpURLConnection;
import java.net.URL;
import org.json.JSONObject;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {



    Button btnFetch;
    TextView txtResult;



    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);



        setContentView(R.layout.activity_main);



        btnFetch = findViewById(R.id.btnFetch);
        txtResult = findViewById(R.id.txtResult);



        btnFetch.setOnClickListener(v -> {
            fetchData();
        });
    }
```

```
   private void fetchData() {



       new Thread(() -> {



           try {



               // Web Service URL

               URL url = new
URL("https://jsonplaceholder.typicode.com/posts/1");



               HttpURLConnection connection =

                       (HttpURLConnection) url.openConnection();


               connection.setRequestMethod("GET");



               connection.setRequestMethod("GET");



               BufferedReader reader = new BufferedReader(new
InputStreamReader(

                               connection.getInputStream()
```

```
        )
);



StringBuilder result = new StringBuilder();



String line;



while ((line = reader.readLine()) != null) {
    result.append(line);
}



reader.close();



// JSON Parsing
JSONObject jsonObject =
        new JSONObject(result.toString());



String title =
        jsonObject.getString("title");
```

```
            String body =
                    jsonObject.getString("body");



            // Display Data
            runOnUiThread(() -> {



                txtResult.setText(
                        "Title: " + title +
                                "\n\nBody: " + body
                );
            });


            connection.disconnect();


        } catch (Exception e) {
            e.printStackTrace();
            runOnUiThread(() -> {
                txtResult.setText(
                        "Error: " + e.getMessage()
                );
            });
        }
```

```
        }).start();

    }

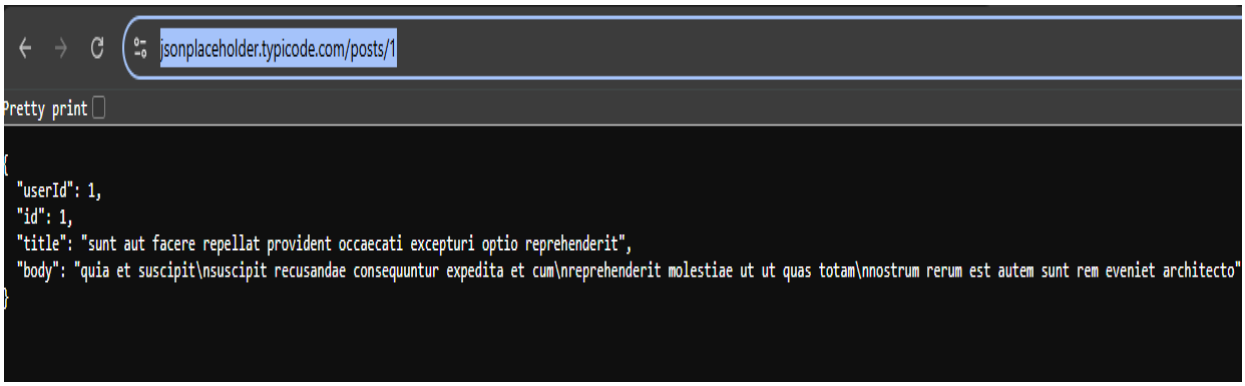
}
```

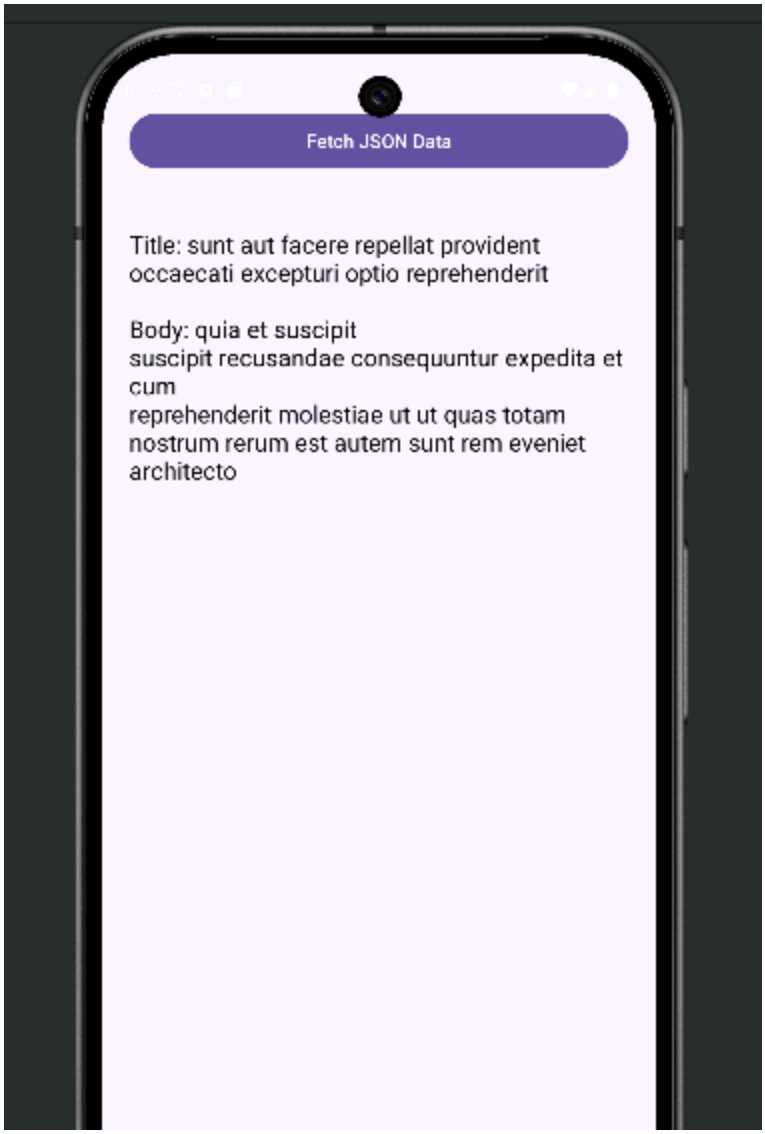
Step 6 : Web Service Used.

API URL : ***https://jsonplaceholder.typicode.com/posts/1***

This is a free test API for students.

https://jsonplaceholder.typicode.com/posts/1




Fetch JSON Data
Title: sunt aut facere repellat provident occaecati excepturi optio reprehenderit
Body: quia et suscipit
suscipit recusandae consequuntur expedita et cum
reprehenderit molestiae ut ut quas totam
nostrum rerum est autem sunt rem eveniet architecto